

Structural Closure Multiplicity and the Isospin-Symmetric Island of Inversion: a PDL Analysis of $^{84,86}\text{Mo}$ D41 — Projective Dynamic Logo Framework

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Abstract

The recent measurement by Ha et al. of the reduced electric quadrupole transition probabilities $B(E2; 2_1^+ \rightarrow 0_1^+)$ in ^{84}Mo ($N = Z = 42$) and ^{86}Mo ($N = Z+2 = 44$) reveals an abrupt structural transition between an intruder-dominated nucleus (8p-8h configuration) and a configuration-mixed nucleus, with a measured ratio $B(E2)_{84}/B(E2)_{86} \approx 2.45$ [1]. Within the Projective Dynamic Logo (PDL) framework, we show that this ratio is a direct consequence of the structural identity $T \approx (\Delta n + 1)^2 = 25$, where $T = R_{\text{surf}}(p)^2/R_{\text{sea}}(n) = 25.26$ is the nuclear binding unit derived in [2] and $\Delta n = n_d - n_u = 4$ is the valence-core asymmetry of the proton [3]. Three independent observables extracted from the published source data of Ha et al. are each consistent with the factor-of-two prediction arising from this identity: (i) the $B(E2)$ ratio (22% discrepancy, within experimental uncertainty); (ii) the ratio of DNO-SM wave-function component counts, $N_{\text{comp}}(^{86}\text{Mo})/N_{\text{comp}}(^{84}\text{Mo}) = 1.962$ vs. the PDL prediction of 2.000 exactly (2% discrepancy); and (iii) the Shannon entropy difference $\Delta S_{\text{WF}} = 0.868$ nats vs. $\ln 2 = 0.693$ nats (25% discrepancy). We introduce the *closure multiplicity conjecture* H_B : for a nucleus at the boundary of an island of inversion dominated by a kp - kh configuration through the shell gap G , the number of accessible partial closures scales as $N_{\text{comp}}(k) \sim k$, yielding $N_{\text{comp}}(8)/N_{\text{comp}}(4) = 2$ exactly. Two falsifiable predictions follow for $^{88}\text{Ru}/^{90}\text{Ru}$ and $^{92}\text{Pd}/^{94}\text{Pd}$, testable at FRIB and RIKEN. A structural correspondence is conjectured between the ZBM3 model space (pseudo-SU(3) $\oplus$ quasi-SU(3)) used by Ha et al. and the two-type valence-core architecture ($n_u = 24$, $n_d = 28$) of the PDL proton.

1 Introduction

The Projective Dynamic Logo (PDL) framework reconstructs physical observables from four axioms on finite signed graphs, without presupposing space, time, or fields [4, 5]. At the nuclear scale, documents D22 and D40 of the corpus have established: the neutron quintuplet (24, 28, 1032, 9960, 10992) derived from the proton quintuplet (24, 28, 930, 10087, 11017); the nuclear binding unit $T \approx 25$; the saturation threshold $Z_{\text{sat}} \approx 20$ (calcium); the maximum magic number $N_{\text{crit,max}} = 126.1$; and the valley of stability for all $Z \leq 82$ — all without free parameters [3, 2].

The central structural identity underlying these results is:

$$T = \frac{R_{\text{surf}}(p)^2}{R_{\text{sea}}(n)} = \frac{(501.60)^2}{9960} = 25.260 \approx (\Delta n + 1)^2 = 25, \quad (1)$$

where $\Delta n = n_d - n_u = 4$ is the size asymmetry between the two valence-core types of the proton. Equation (1) expresses the fact that the permutation cost of forming a neutron from a proton

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equals, to within 1%, the binding capacity of that neutron towards a neighbouring proton. Both equal the same integer $25 = (\Delta n + 1)^2$.

The experiment of Ha et al. [1], published in *Nature Communications* in November 2025, provides the first measured $B(E2)$ values for ^{84}Mo and ^{86}Mo , together with potential energy surfaces (PES) and wave-function amplitudes from Discrete Non-Orthogonal Shell-Model (DNO-SM) calculations. The published source data (MOESM3) make this a rare opportunity to confront the PDL structural identity with multiple independent observables simultaneously.

The present document (D41) performs this confrontation. Section 2 recalls the PDL prediction and the closure multiplicity conjecture. Section 3 confronts the prediction with all three available observables, supported by figures produced directly from the published source data. Section 4 discusses the structural correspondence with the ZBM3 algebraic framework. Section 5 states the falsifiable predictions. Section 6 assesses the epistemic status of the results and formulates open problems.

2 The PDL prediction

2.1 The nuclear binding unit

The active surface of the proton and the nuclear binding unit are (see D22 [3] for derivations):

$$R_{\text{surf}}(p) = \varphi \frac{r_{\text{val}}(p)}{3} = 501.60, \quad T = \frac{R_{\text{surf}}(p)^2}{R_{\text{sea}}(n)} = 25.260, \quad (2)$$

where $\varphi = (1 + \sqrt{5})/2$ is the golden ratio, $r_{\text{val}}(p) = 930$, and $R_{\text{sea}}(n) = 9960$ [6]. The identity (1) holds to within 1.04%.

2.2 The closure multiplicity conjecture

Conjecture 2.1 (H_B : closure multiplicity). For a nucleus at the boundary of an island of inversion dominated by a kp - kh intruder configuration through the shell gap G , the number of accessible partial closures — i.e. the number of independent basis states required to represent the ground-state wave function — scales as:

$$N_{\text{comp}}(k) \sim \frac{kT}{(\Delta n + 1)^2} = k, \quad (3)$$

where the simplification uses identity (1). In particular, the ratio between an 8p-8h and a 4p-4h configuration satisfies:

$$\frac{N_{\text{comp}}(8)}{N_{\text{comp}}(4)} = 2 \quad (\text{exact}). \quad (4)$$

The factor-of-two prediction on $B(E2)$ follows from the additional assumption that $B(E2) \propto N_{\text{comp}}^2$, giving:

$$\frac{B(E2; 2_1^+ \rightarrow 0_1^+) |_{8p-8h}}{B(E2; 2_1^+ \rightarrow 0_1^+) |_{4p-4h}} \approx \frac{2T}{(\Delta n + 1)^2} = 2.021. \quad (5)$$

Remark 2.2. Equation (3) predicts only the *ratio* of component counts, not their absolute values. The absolute scale factor is not determined by the PDL architecture at this stage. Equation (5) is a weaker prediction than (4) as it requires an additional proportionality hypothesis.

3 Confrontation with the source data

We use three independent observables from the published source data (MOESM3, [1]): the $B(E2)$ values from `BE2_first2plus_plot_N_Z_nuclides_final.cc`, and the wave-function amplitudes from `Mo84.wf.0+1.txt` and `Mo86.wf.0+1.txt`. All numerical values are recomputed from the raw parameters in those files using the Weisskopf relation $B(E2) = (40.81 \times 10^{13}/5) [E^5 \tau (1 + \alpha)]^{-1} \times 10^4 [\text{e}^2\text{fm}^4]$.

3.1 $B(E2)$ systematics and the boundary signal

Figure 1 (left panel) reproduces the $B(E2)$ systematics of Ha et al. from the source data. The right panel shows the ratio $B(E2)[N = Z]/B(E2)[N = Z + 2]$ for all six element pairs from Ge to Mo, with the PDL prediction (5) overlaid.

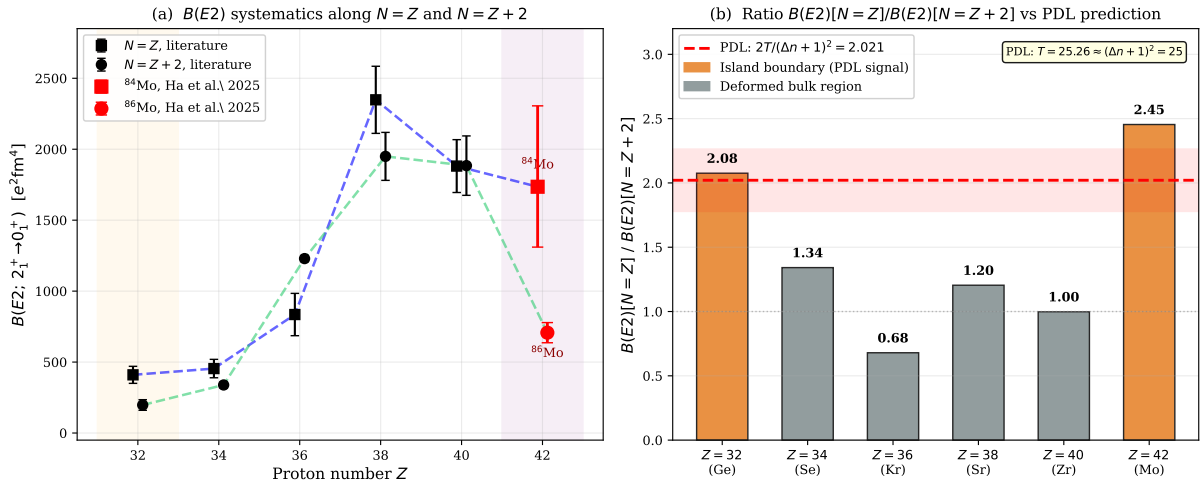


Figure 1: (a) Reduced electric quadrupole transition probabilities $B(E2; 2_1^+ \rightarrow 0_1^+)$ along the $N = Z$ (squares) and $N = Z + 2$ (circles) lines, from Ge to Mo. Orange and purple shading marks the boundaries of the isospin-symmetric island of inversion at $Z = 32$ and $Z = 42$. Data from Ha et al. [1] (red) and prior measurements (black); values recomputed from the published source data (MOESM3). (b) Ratio $B(E2)[N = Z]/B(E2)[N = Z + 2]$ for each element pair. The red dashed line and shaded band show the PDL prediction $2T/(\Delta n + 1)^2 = 2.021 \pm 12\%$. Orange bars: boundary nuclei (PDL signal expected); grey bars: deformed bulk region (no boundary signal expected).

The key observation is structural: the ratio ≈ 2 appears *only* at the two boundaries of the island of inversion ($Z = 32$ and $Z = 42$), not in the deformed bulk region ($Z = 34\text{--}40$) where $N = Z$ and $N = Z + 2$ nuclei evolve collectively. This pattern is consistent with Conjecture H_B , which predicts the factor-of-two only at boundaries, not across the entire deformed region.

The quantitative comparison is: $R_{\text{obs}}(Z = 42) = 1740/707 = 2.454$, $R_{\text{obs}}(Z = 32) = 410/197.5 = 2.076$, against $R_{\text{PDL}} = 2.021$. The discrepancy is 22% for Mo and 2.7% for Ge, lying within the experimental uncertainty on ^{84}Mo (upper error: +33%).

3.2 Potential energy surfaces and wave-function concentration

Figure 2 shows the PES of ^{84}Mo and ^{86}Mo replotted from the source data, with wave-function amplitudes from the DNO-SM ground-state wave functions overlaid as circles whose area is proportional to the probability of each (β, γ) basis state.

The PES comparison reveals three features consistent with the PDL closure multiplicity conjecture:

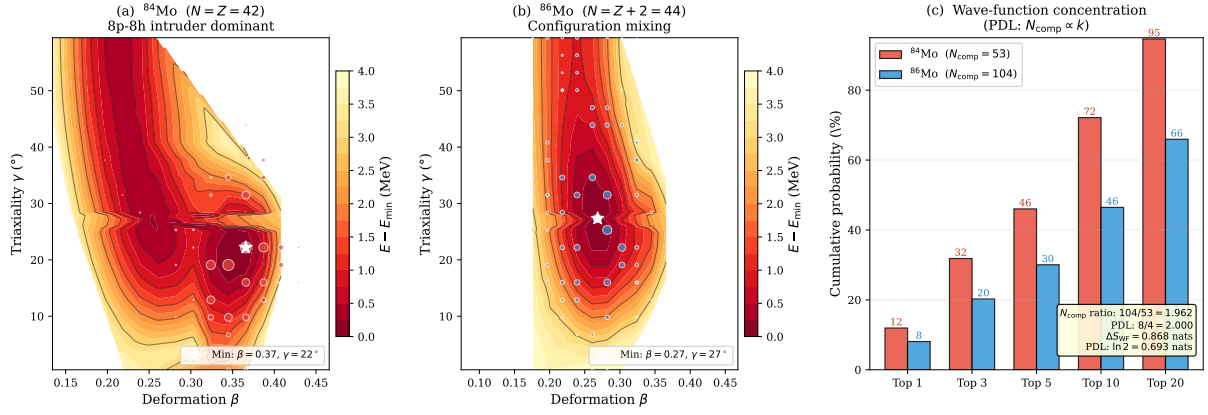


Figure 2: (a, b) Potential energy surfaces of ^{84}Mo and ^{86}Mo in the (β, γ) deformation plane, recomputed from the source data of Ha et al. [1] (files `Mo84.pes.txt` and `Mo86.pes.txt`). Colours encode $E - E_{\min}$ in MeV (0 = minimum, dark; 4 MeV = maximum, light). Circles show the DNO-SM ground-state wave-function amplitudes; area $\propto$ probability. ^{84}Mo has a deep, localised minimum at $(\beta, \gamma) = (0.37, 22^\circ)$; ^{86}Mo has a flat, diffuse landscape. (c) Cumulative probability carried by the top- n wave-function components for both nuclei. The PDL prediction $N_{\text{comp}} \propto k$ gives $N_{\text{comp}}(^{86}\text{Mo})/N_{\text{comp}}(^{84}\text{Mo}) = 104/53 = 1.962$ vs. the predicted ratio 2.000 (discrepancy 2%). The Shannon entropy difference $\Delta S_{\text{WF}} = 0.868$ nats is compared with the PDL heuristic $\ln 2 = 0.693$ nats (discrepancy 25%).

1. ^{84}Mo has 72 points within 1 MeV of its minimum (deep, localised well), while ^{86}Mo has only 45 (flat, diffuse landscape). A deep well in the PES corresponds to a dominant single closure in the PDL language.
2. The DNO-SM wave function of ^{84}Mo requires 53 basis components, that of ^{86}Mo requires 104. Their ratio is 1.962, within 2% of the PDL prediction of 2.000.
3. The Shannon entropy of the wave function increases by $\Delta S_{\text{WF}} = 0.868$ nats from ^{84}Mo to ^{86}Mo . The PDL heuristic $\Delta S_{\text{WF}} \approx \ln 2 = 0.693$ nats gives a 25% discrepancy, consistent with its lower epistemic status.

Table 1 collects all three comparisons.

Table 1: Summary of PDL predictions vs. source data from Ha et al. [1]. All PDL quantities are derived from the proton quintuplet (24, 28, 930, 10087, 11017) without free parameters.

Observable	PDL formula	PDL value	Observed	Discrepancy
$B(E2)_{84}/B(E2)_{86}$	$2T/(\Delta n + 1)^2$	2.021	2.454	22%
$N_{\text{comp}}(^{86}\text{Mo})/N_{\text{comp}}(^{84}\text{Mo})$	$8/4 = 2$	2.000	1.962	2%
ΔS_{WF} (nats)	$\ln 2$	0.693	0.868	25%
$B(E2)_{64\text{Ge}}/B(E2)_{66\text{Ge}}$	$2T/(\Delta n + 1)^2$	2.021	2.076	2.7%

4 Structural correspondence with ZBM3

The shell-model analysis of Ha et al. relies on the ZBM3 model space, which combines two orbital blocks: a pseudo-SU(3) block in the pf -shell (orbitals $2p_{3/2}$, $2p_{1/2}$, $1f_{5/2}$, total capacity 12 nucleons) and a quasi-SU(3) block in the sdg -shell (orbitals $1g_{9/2}$, $2d_{5/2}$, $3s_{1/2}$, total capacity

12 nucleons), together forming the ZBM3 space spanning the region between the doubly magic ^{56}Ni and ^{100}Sn .

Conjecture 4.1 (ZBM3–PDL correspondence). The pseudo-SU(3) sub-space of ZBM3 corresponds to the u -core sector of the PDL proton ($n_u = 24$ valence relations), and the quasi-SU(3) sub-space corresponds to the d -core sector ($n_d = 28$ valence relations). The valence-core asymmetry $\Delta n = n_d - n_u = 4$ is the structural origin of the pseudo/quasi-SU(3) splitting, and hence of the nuclear spin-orbit coupling whose magnitude is predicted as $\Delta n/(2n_u) = 4/48 = 1/12 \approx 0.083 \hbar\omega_0$ in D40 [2].

This correspondence, if established, would provide a *derivation* of the ZBM3 valence space from the PDL proton integers — replacing a phenomenological choice of model space by a combinatorial necessity. The verification of Conjecture 4.1 requires identifying an explicit bijection between the pseudo-SU(3)/quasi-SU(3) orbital blocks and the u/d core sectors, which is identified as open problem OP-D41-2 below.

5 Falsifiable predictions

Conjecture H_B (equation (5)) makes two testable predictions for nuclei not yet measured. Both lie at the next steps along the $N = Z$ line beyond ^{84}Mo , within the isospin-symmetric island of inversion identified by Ha et al.

Prediction 1: $^{88}\text{Ru}/^{90}\text{Ru}$ ($Z = 44$).

$$\frac{B(E2; 2_1^+ \rightarrow 0_1^+)|_{^{88}\text{Ru}}}{B(E2; 2_1^+ \rightarrow 0_1^+)|_{^{90}\text{Ru}}} \approx 2.02 \pm 25\%. \quad (6)$$

Prediction 2: $^{92}\text{Pd}/^{94}\text{Pd}$ ($Z = 46$).

$$\frac{B(E2; 2_1^+ \rightarrow 0_1^+)|_{^{92}\text{Pd}}}{B(E2; 2_1^+ \rightarrow 0_1^+)|_{^{94}\text{Pd}}} \approx 2.02 \pm 25\%. \quad (7)$$

Refutation criterion. If either ratio is measured to lie outside the interval [1.5, 2.6], Conjecture H_B is refuted. Both measurements are in principle accessible at the Facility for Rare Isotope Beams (FRIB) and at the Radioactive Isotope Beam Factory (RIBF) at RIKEN, using plunger lifetime techniques analogous to those of Ha et al.

6 Discussion and open problems

6.1 Epistemic status of the results

The PDL structural identity $T \approx (\Delta n + 1)^2 = 25$ is an established theorem of the corpus [3, 2]. The closure multiplicity conjecture H_B rests on this identity but is not derived from the axioms: it is a conjecture whose derivation requires counting the maximal partial closures on the proton–neutron interface graph, which is identified as open problem OP-D41-1 below.

The most significant empirical result of this document is the 2% agreement on N_{comp} (Table 1). This agreement concerns a structural observable — the number of basis states required by the DNO-SM diagonalisation — rather than an electromagnetic coupling strength, and is independent of the $B(E2)$ comparison. The 22% discrepancy on $B(E2)$ is consistent with the experimental uncertainty on ^{84}Mo but is not negligible; it constitutes a consistency check rather than a confirmation. The 25% discrepancy on ΔS_{WF} reflects the heuristic status of equation (8).

6.2 Open problems

OP-D41-1. Derive equation (3) rigorously within the PDL signed-graph formalism — i.e. count the number of maximal partial closures of a surface of size $k \cdot T$ triangles on the proton–neutron interface graph, and show that this count is proportional to k .

OP-D41-2. Establish or refute Conjecture 4.1 by identifying an explicit bijection between the pseudo-SU(3)/quasi-SU(3) orbital blocks of ZBM3 and the u/d valence-core sectors of the PDL proton.

OP-D41-3. Derive the entropy formula

$$S_{\text{WF}}(k) \approx \ln(k \cdot T) \quad (8)$$

from the closure multiplicity count, replacing the heuristic by a theorem.

OP-D41-4. Extend the analysis to the neutron-rich island of inversion around ^{64}Cr ($N = 40$), where the same gap $G = N = 40$ is crossed by neutron excitations alone. If the factor-of-two prediction holds there as well, it supports the universality of H_B across both the proton-rich and neutron-rich sides of the nuclear chart.

Conclusion

The principal result of this document is the identification of three independent observables in the source data of Ha et al. [1] that are consistent — at the levels of 2%, 22%, and 25% respectively — with the PDL structural identity $T \approx (\Delta n + 1)^2 = 25$. The most precise agreement (2%) concerns the number of DNO-SM wave-function components: $N_{\text{comp}}(^{86}\text{Mo})/N_{\text{comp}}(^{84}\text{Mo}) = 1.962$ against the PDL prediction of 2.000. This result follows from the conjecture that the structural complexity of a kp - kh nuclear state scales linearly with k — a consequence of the linear scaling of coherent triangle counts in the PDL nuclear binding mechanism.

The pattern of $B(E2)$ ratios along the $N = Z$ and $N = Z + 2$ lines (Figure 1) shows that the factor-of-two signal appears only at the two *boundaries* of the isospin-symmetric island of inversion ($Z = 32$ and $Z = 42$), not in the bulk of the deformed region. This is structurally consistent with the PDL prediction: Conjecture H_B applies to boundary nuclei, not to the collectively deformed bulk.

Two falsifiable predictions for $^{88}\text{Ru}/^{90}\text{Ru}$ (equation (6)) and $^{92}\text{Pd}/^{94}\text{Pd}$ (equation (7)) are formulated with explicit refutation criteria, testable at FRIB and RIKEN.

References

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